SIT-REP FILTER-WHEEL

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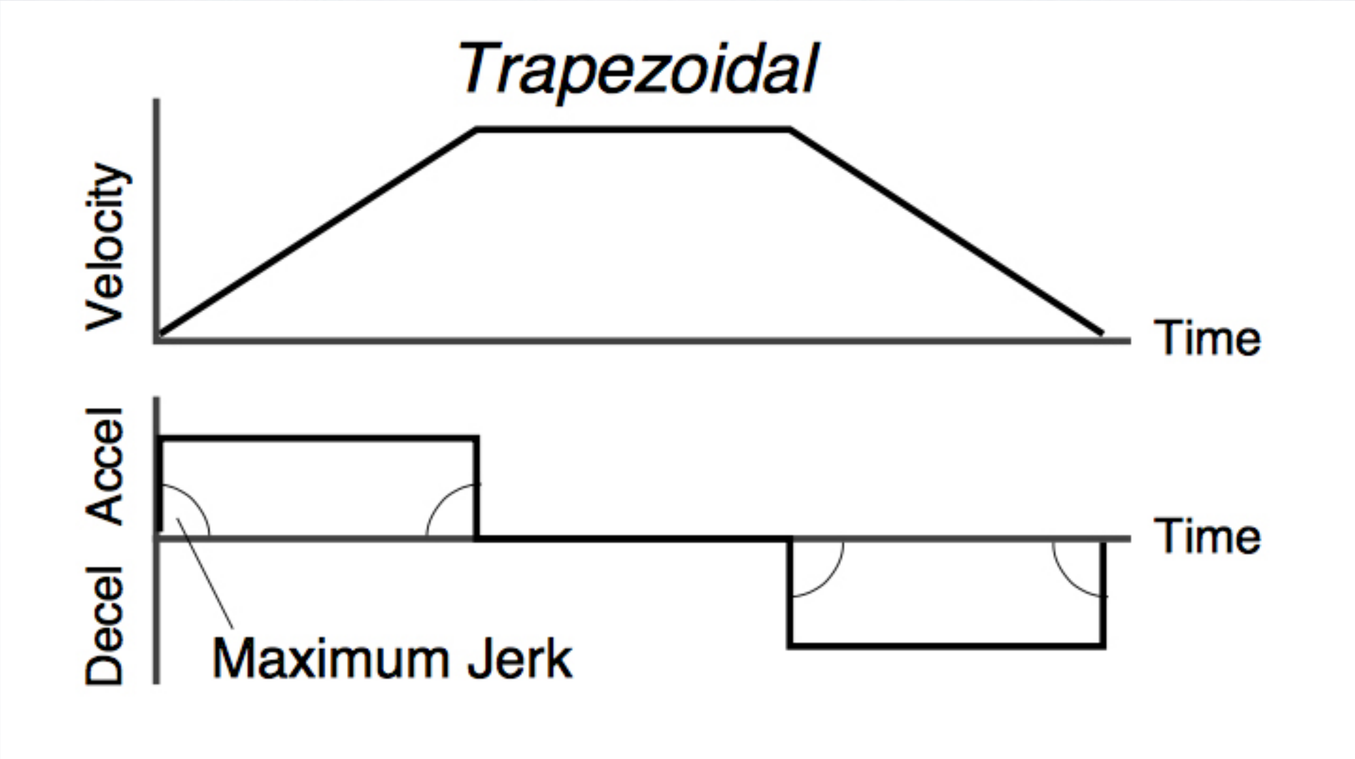
OVERVIEW

- Position Curves
  - Trapezoidal Model
  - S-Curve Model
- VHDL Testing on Spartan Board

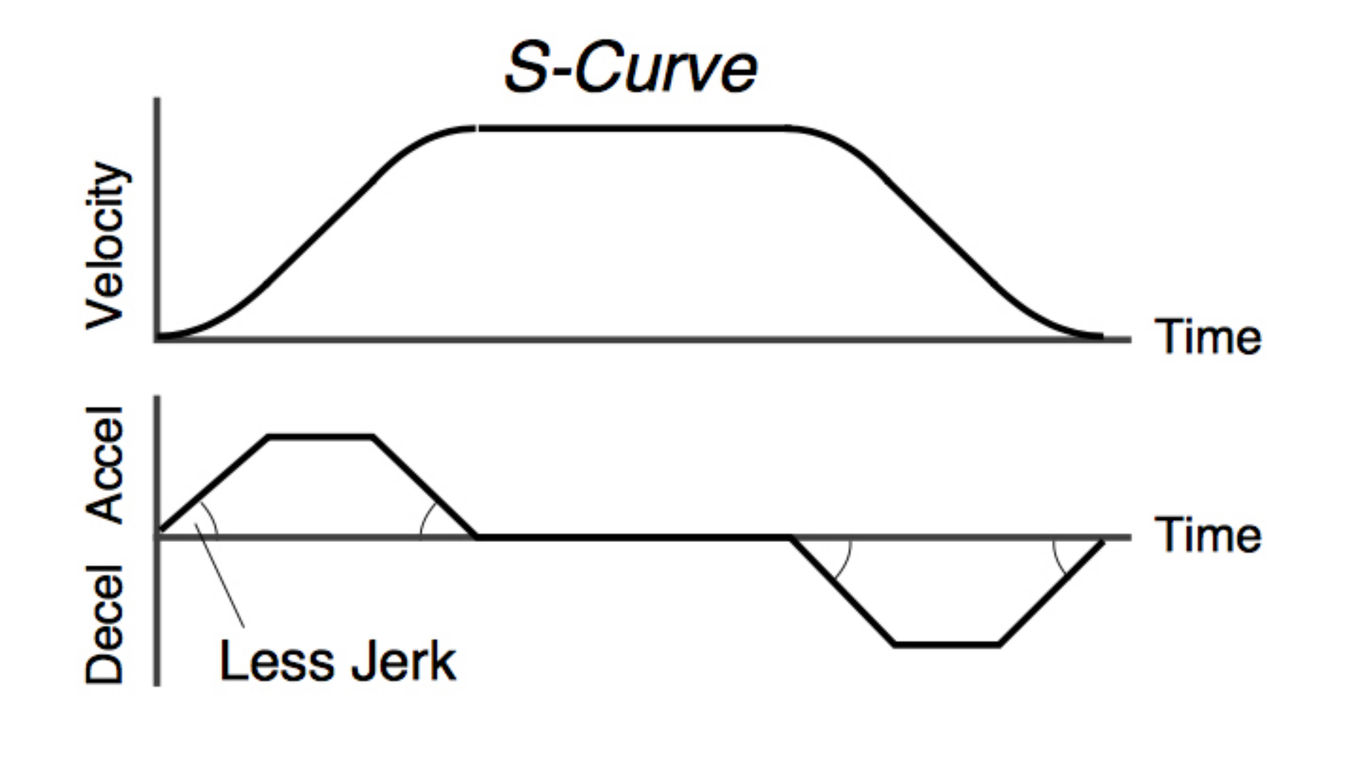
POSITION CURVE

- Start with Acceleration Curve
- Integrate to get velocity
- Integrate to get position
- Use position to control motor

TRAPEZOIDAL PROFILE



S-CURVE PROFILE



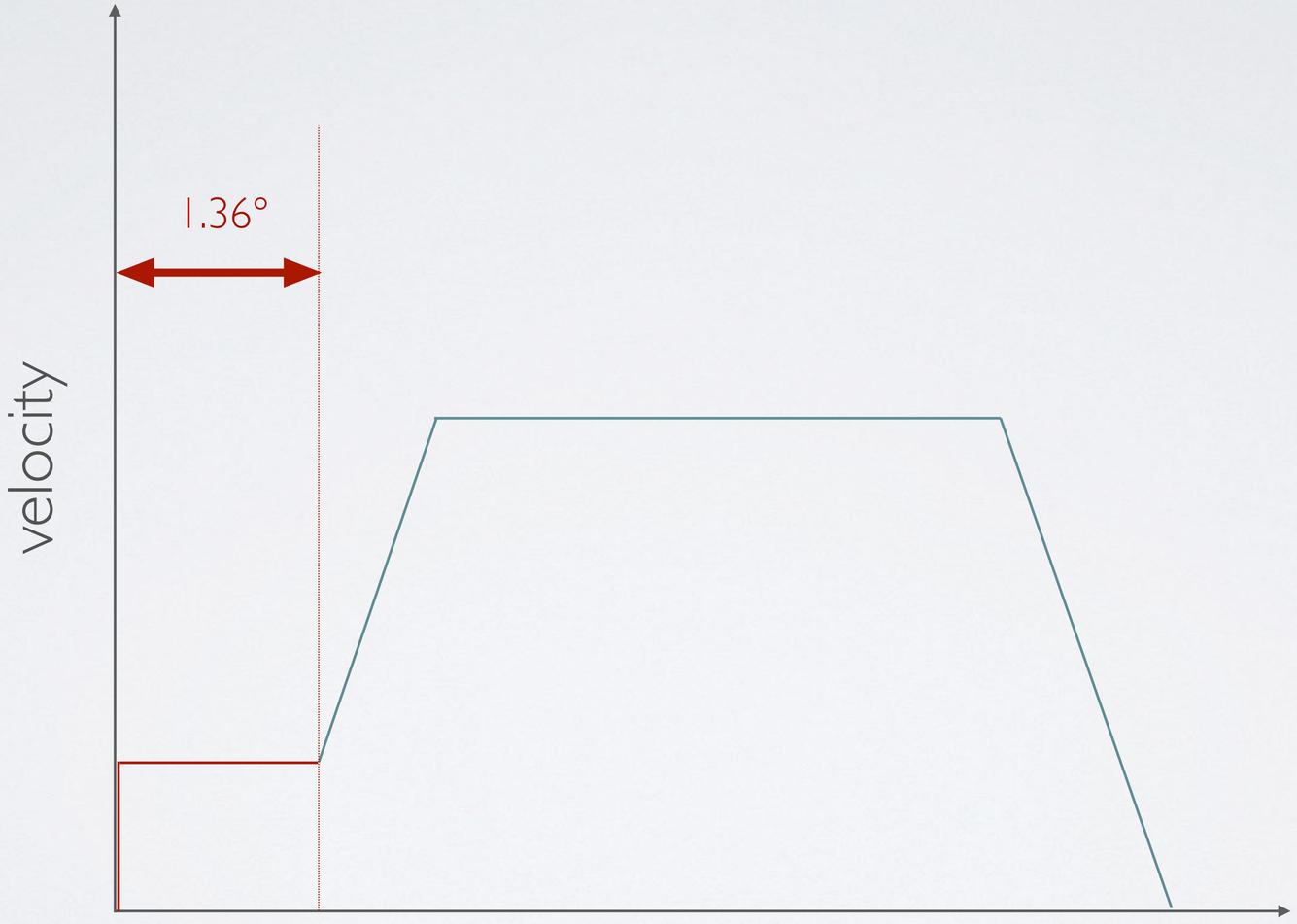
S-CURVE

+ less jerk/vibrations
+ less wear/tear
- more complex
- slower

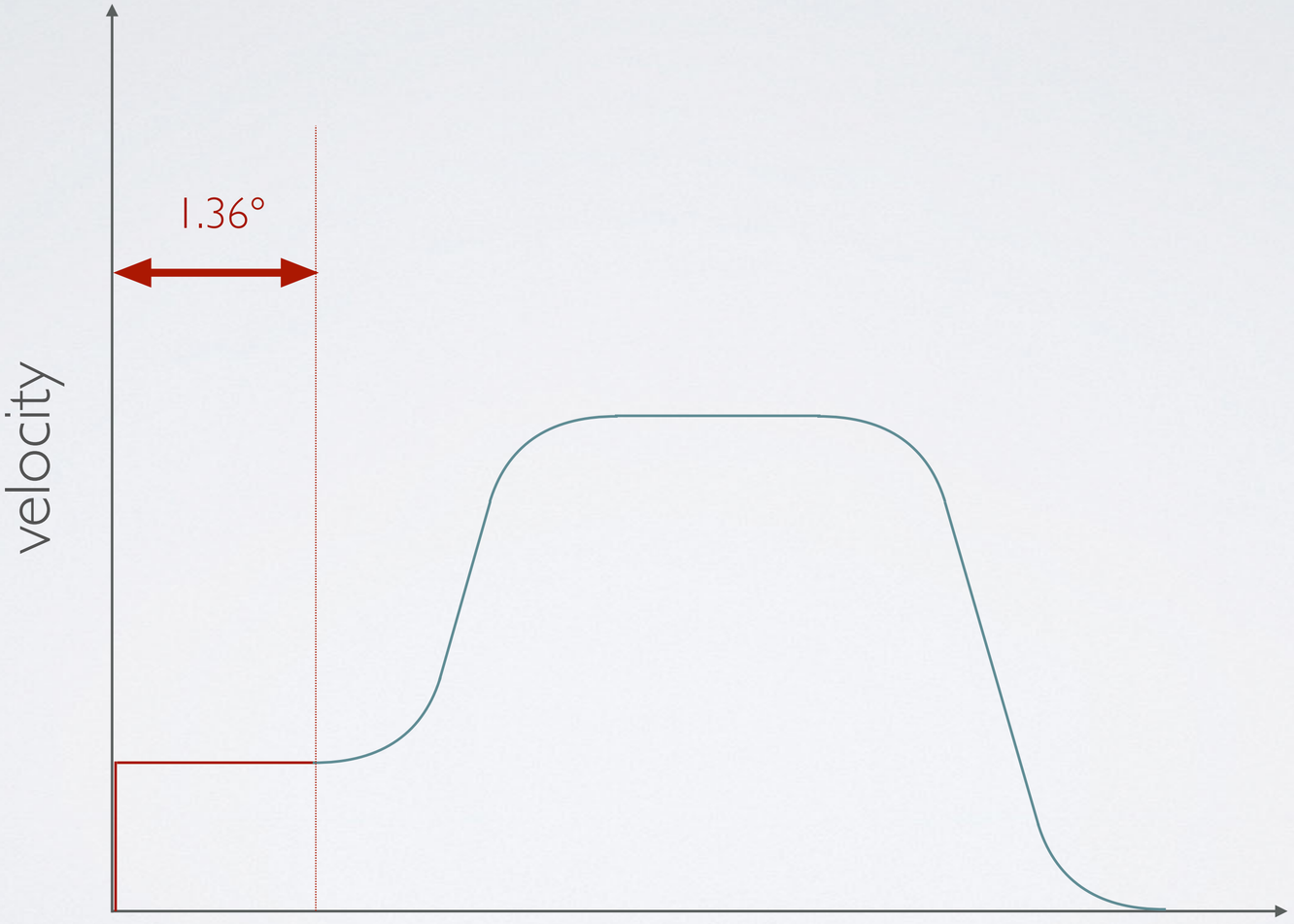
TRAPEZOIDAL

- more jerk/vibrations
- more wear/tear
+ simple
+ faster

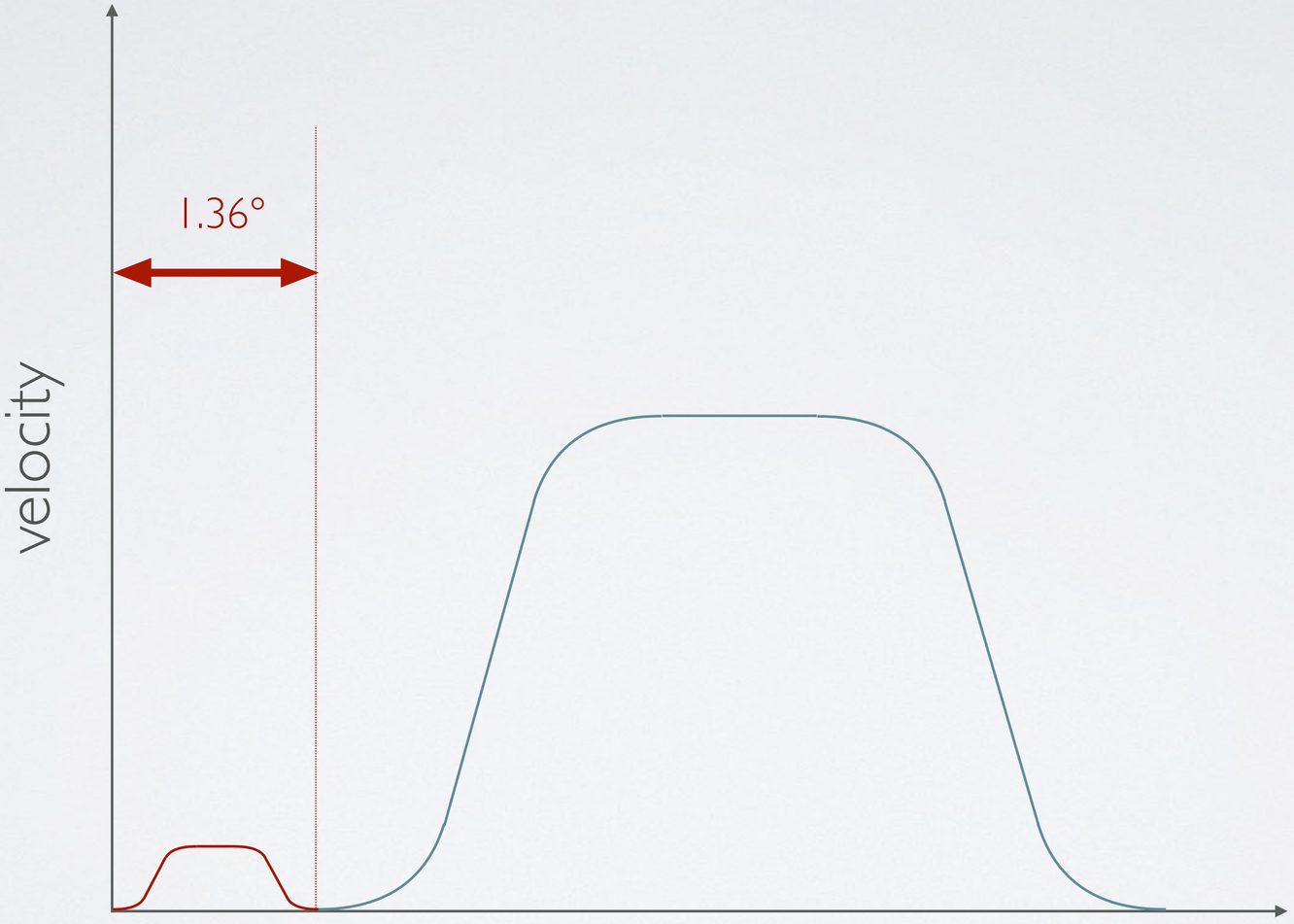
BACKLASH, TRAPEZOIDAL



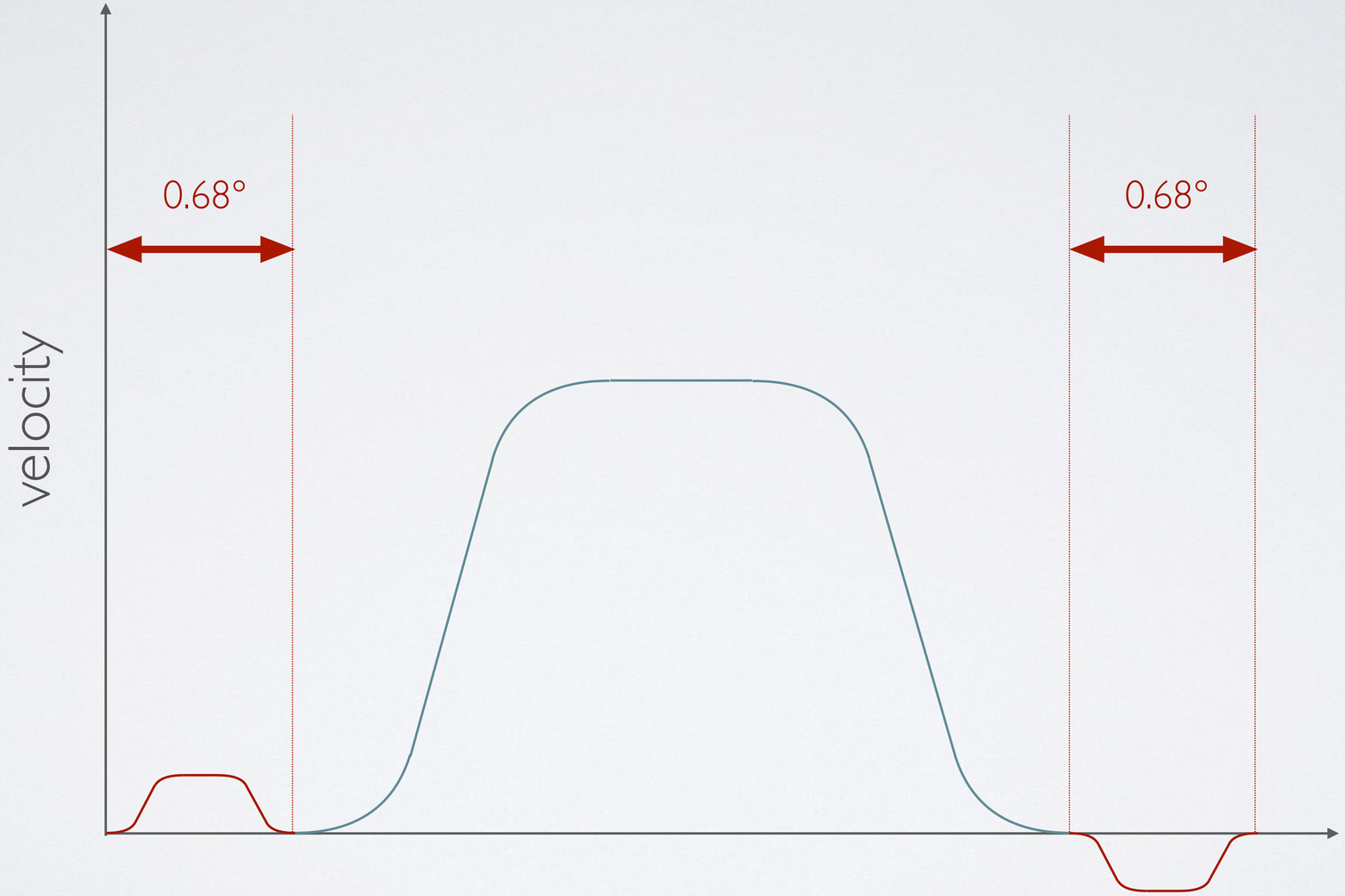
BACKLASH, S-CURVE



BACKLASH, S-CURVE



BACKLASH, S-CURVE

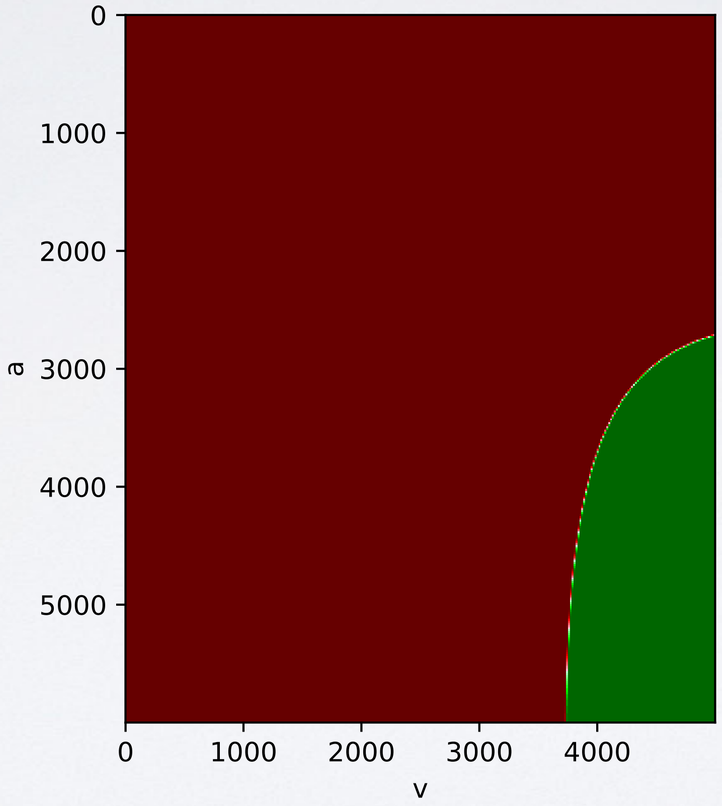


PARAMETERS

- Max Velocity
- Max Acceleration
- Max Jerk (Steepness of acceleration)
- Final Position
- Maximum Time (0.5s)

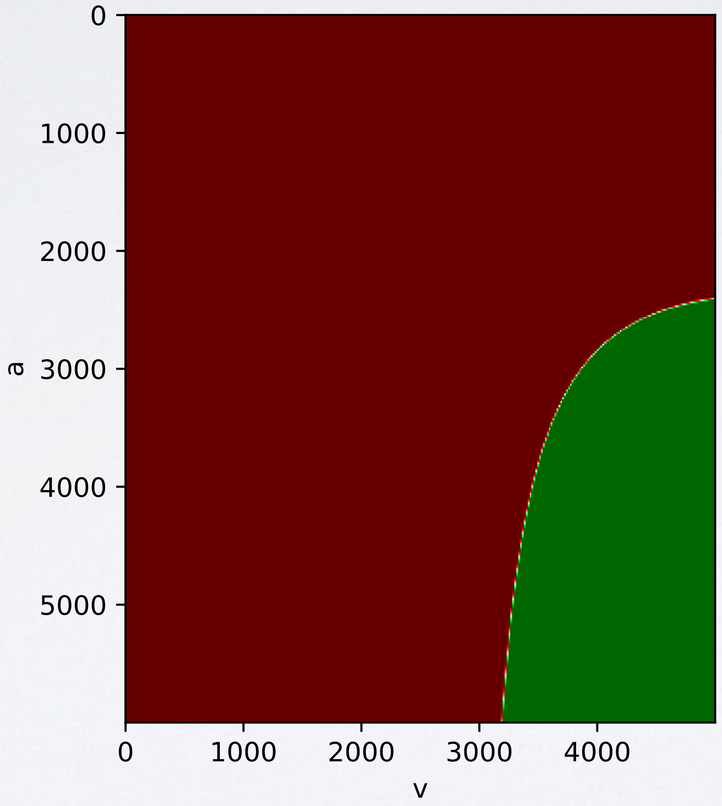
PARAMETER SPACE

m = 100000 rad/s^3, angle = 105°



PARAMETER SPACE

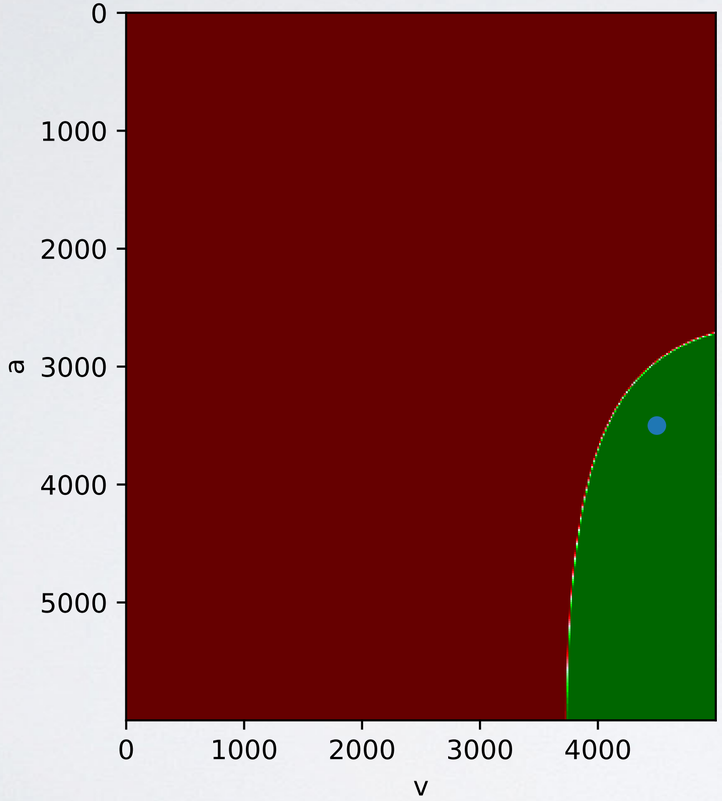
m = 1000000 rad/s^3, angle = 105°

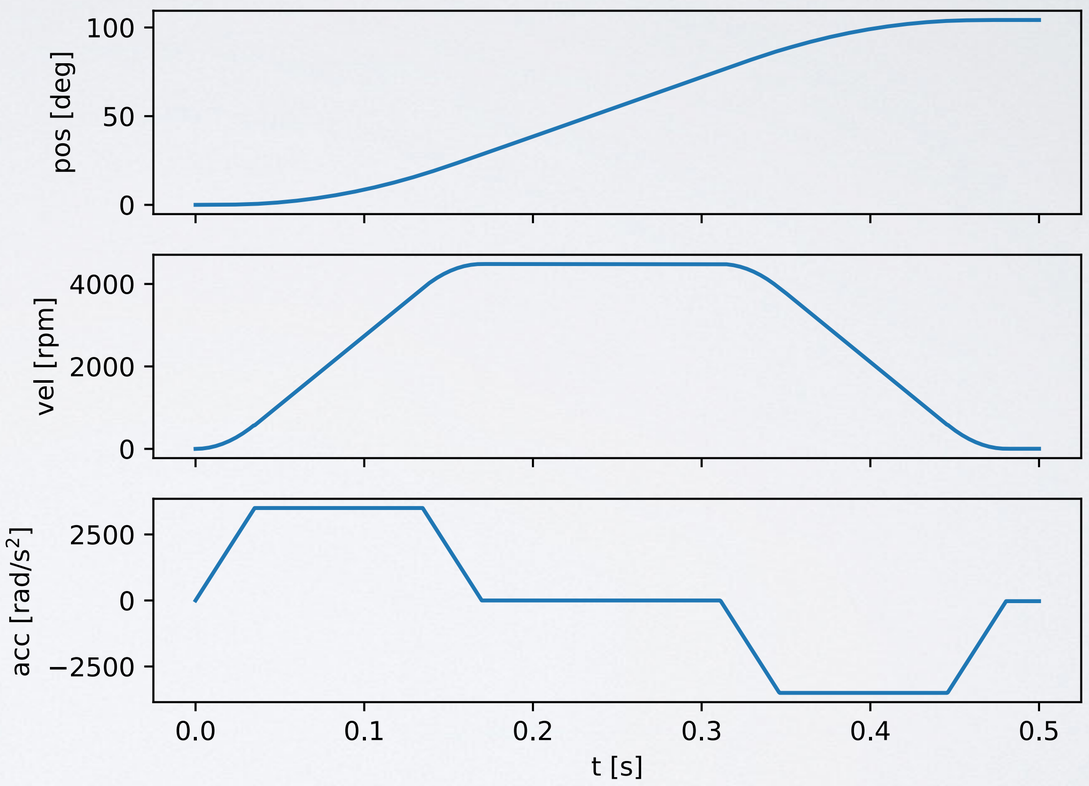


m = 1000000 rad/s^3 is really high, closer to trapezoidal than S-curve!

PARAMETER SPACE

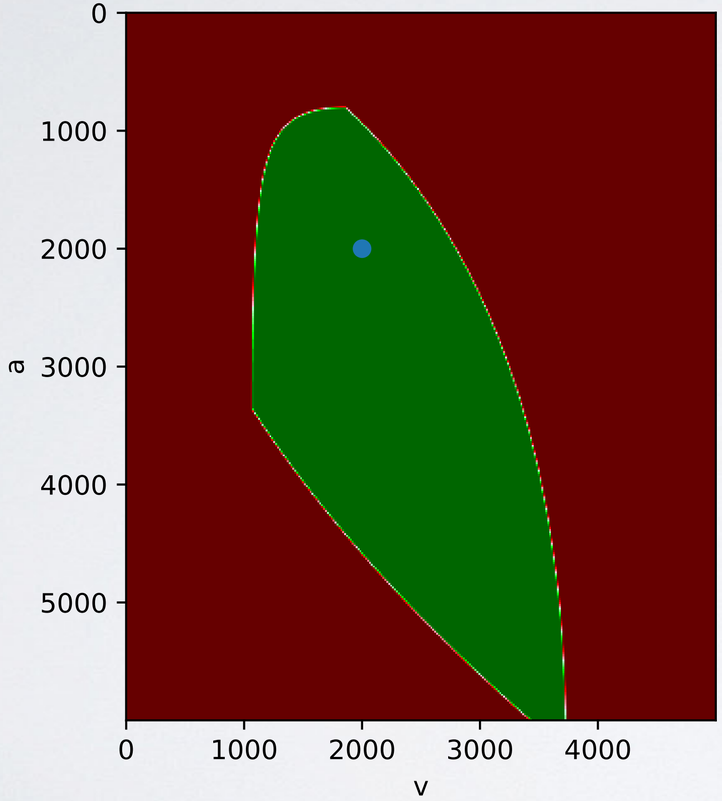
m = 100000 rad/s^3, angle = 105°

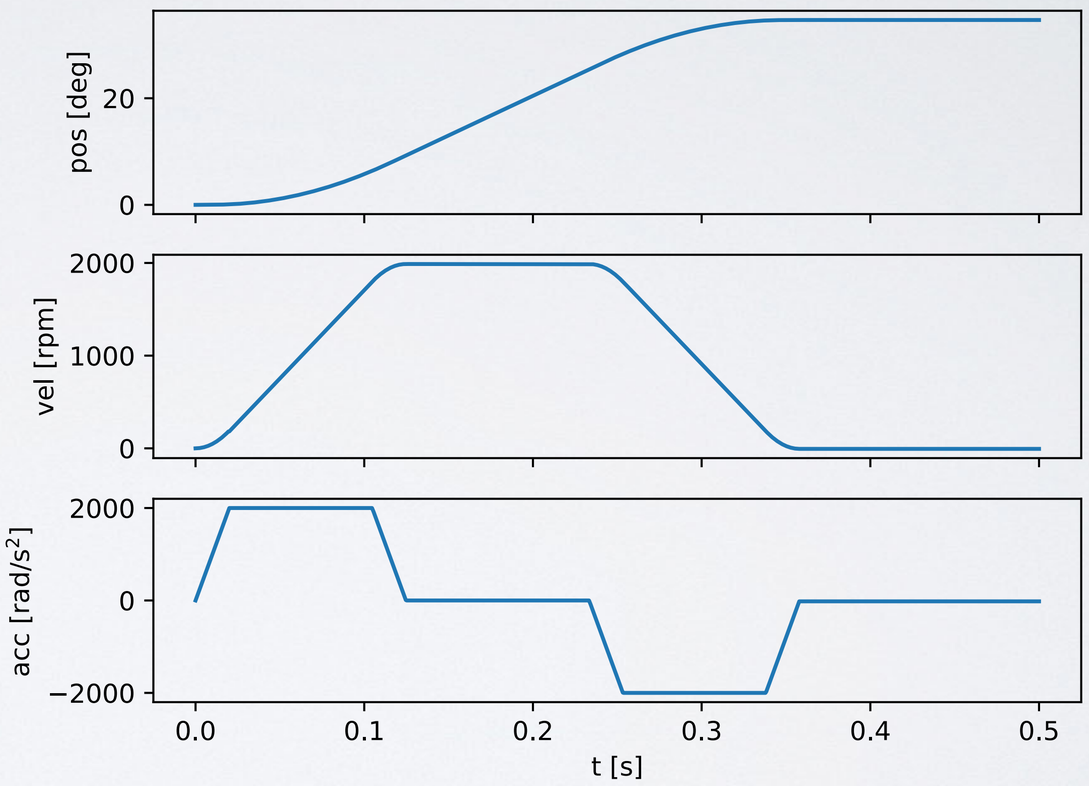




PARAMETER SPACE

m = 100000 rad/s^3, angle = 35°





VHDL TESTING ON SPARTAN BOARD

- Installed Software and Drivers on Virtual Machine (Ubuntu)
  - Working on introductory VHDL-Tutorials
  - LED, Buttons Test

FUTURE

- Adapt Curves to discretised FPGA architecture
- Understand old NIM-VHDL Code from Johannes
- Build first Model in VHDL (use on-board LEDs instead of driving the motor at first)
- Adapt curves further:
  - Asymmetric Movement: faster deceleration than acceleration
  - Include Backlash-Compensation

FUTURE

Today

Portugal

Mo	Di	Mi	Do	Fr	Sa	So	Mo	Di	Mi	Do	Fr	Sa
12.10.	13.10.	14.10.	15.10.	16.10.	17.10.	18.10.	19.10.	20.10.	21.10.	22.10.	23.10.	24.10.

Quarantine in ZH

Mo	Di	Mi	Do	Fr	Sa	So	Mo	Di	Mi	Do	Fr	Sa
25.10.	26.10.	27.10.	28.10.	29.10.	30.10.	31.10.	1.11.	2.11.	3.11.	4.11.	5.11.	6.11.